

YUMENZHEN, China

WMO: 524360

Lat: 40.2695N

Lon: 97.0271E

Elev: 5010

StdP: 12.22

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.5	(c) -0.1	(d) -18.3	(e) 2.4	(f) 10.4	(g) -14.7	(h) 3.0	(i) 13.3	(j) 23.3	(k) 21.5	(l) 20.9	(m) 21.1	(n) 5.4	(o) 90	(p) 0.525

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 23.2	(c) 90.0	(d) 60.8	(e) 87.6	(f) 59.7	(g) 85.3	(h) 58.8	(i) 64.3	(j) 80.9	(k) 62.7	(l) 79.9	(m) 61.4	(n) 78.2	(o) 7.3	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.6	92.0	68.3	57.7	85.6	67.7	55.6	79.4	67.9	32.3	81.1	31.0	79.4	29.9	78.6	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	18.5	16.2	-10.7	94.6	5.0	2.2	-14.3	96.2	-17.2	97.5	-20.0	98.7	-23.6	100.3
			-11.4	67.6	4.7	2.4	-14.8	69.4	-17.6	70.8	-20.2	72.1	-23.7	73.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	46.9	15.6	24.3	37.4	51.2	61.1	69.1	72.6	70.0	60.5	47.3	33.1	19.0	(6)
(7)		DBStd	20.84	7.57	7.97	8.82	8.29	6.23	4.59	4.34	4.52	5.79	6.74	7.91	7.66	(7)
(8)		HDD50	3870	1067	719	395	83	5	0	0	0	3	130	508	960	(8)
(9)		HDD65	7208	1532	1139	855	417	149	17	2	11	156	547	958	1425	(9)
(10)		CDD50	2729	0	0	5	118	348	572	701	621	316	48	0	0	(10)
(11)		CDD65	594	0	0	0	3	27	139	238	167	20	0	0	0	(11)
(12)		CDH74	7083	0	0	0	67	482	1596	2675	1931	328	4	0	0	(12)
(13)		CDH80	2283	0	0	0	10	78	452	1044	649	50	0	0	0	(13)
(14)	Wind	WSAvg	7.1	7.7	7.8	8.5	8.7	6.9	6.0	5.7	5.8	5.8	6.2	7.8	8.0	(14)
(15)	Precipitation	PrecAvg	2.3	0.0	0.1	0.2	0.2	0.3	0.4	0.6	0.4	0.2	0.1	0.1	0.0	(15)
(16)		PrecMax	3.7	0.2	0.3	0.6	0.7	1.6	1.5	2.7	1.4	0.8	0.6	0.4	0.2	(16)
(17)		PrecMin	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	0.8	0.0	0.1	0.2	0.2	0.4	0.4	0.6	0.4	0.2	0.1	0.1	0.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.5	51.1	66.8	80.6	85.5	90.2	93.9	92.5	85.0	74.2	59.0	43.3	(19)
(20)			MCWB	29.3	34.2	42.6	50.0	54.4	59.2	62.9	61.3	56.3	49.1	40.0	30.4	(20)
(21)		2%	DB	35.7	45.6	61.8	74.8	82.3	87.3	90.9	89.0	81.0	69.5	54.6	38.2	(21)
(22)			MCWB	25.8	31.5	40.4	47.6	53.1	58.1	61.7	60.1	54.3	47.5	38.0	27.7	(22)
(23)		5%	DB	31.5	42.2	57.5	70.9	79.4	85.0	88.5	86.3	77.9	65.5	50.8	34.2	(23)
(24)			MCWB	23.4	29.8	38.4	45.8	51.6	57.4	61.2	59.1	53.3	45.5	36.1	25.5	(24)
(25)		10%	DB	27.8	38.5	53.5	67.4	76.4	82.6	86.0	83.8	74.8	61.7	46.8	30.7	(25)
(26)			MCWB	21.1	27.7	36.5	44.2	50.6	56.8	60.3	58.6	52.5	43.4	34.1	23.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.6	34.9	43.3	51.4	56.0	63.7	68.2	66.1	60.0	51.1	41.4	31.0	(27)
(28)			MADB	40.9	49.4	65.0	77.1	81.0	79.5	86.4	81.5	72.1	66.7	56.4	42.2	(28)
(29)		2%	WB	26.1	32.0	40.9	48.5	54.3	61.8	65.2	63.6	57.1	48.4	38.5	28.0	(29)
(30)			MADB	35.3	44.7	60.3	72.4	77.9	78.5	82.6	79.5	73.5	67.3	53.3	37.3	(30)
(31)		5%	WB	23.6	30.1	38.9	46.6	53.0	60.3	63.5	61.9	55.4	46.4	36.4	25.8	(31)
(32)			MADB	31.2	41.5	56.5	68.8	75.5	77.3	81.0	78.4	72.7	64.0	49.9	33.6	(32)
(33)		10%	WB	21.2	28.1	36.8	44.8	51.5	58.9	62.1	60.6	53.8	44.2	34.4	23.8	(33)
(34)			MADB	27.4	38.0	52.6	65.4	73.3	76.7	79.6	77.4	70.7	60.3	46.4	30.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	20.1	23.2	24.6	24.3	24.0	23.0	23.2	24.3	25.1	24.6	20.2	18.0	(35)
(36)			MCDBR	24.2	27.6	30.0	29.3	29.4	28.1	28.6	29.8	29.4	29.3	24.9	21.9	(36)
(37)			MCWBR	17.0	17.1	15.8	13.1	12.5	11.3	10.5	10.3	11.2	14.0	14.0	14.6	(37)
(38)		5% WB	MCDBR	23.7	26.0	28.3	26.7	25.6	22.7	22.9	23.6	23.7	27.5	23.9	21.1	(38)
(39)			MCWBR	16.7	16.3	15.0	12.0	11.1	9.9	9.6	9.0	9.5	13.3	13.7	14.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.300	0.350	0.411	0.465	0.466	0.480	0.468	0.449	0.420	0.372	0.328	0.300	(40)	
(41)		taud	2.208	2.071	1.920	1.821	1.876	1.910	2.001	2.031	2.047	2.090	2.158	2.209	(41)	
(42)		Ebn at Noon	269	268	262	257	259	256	257	259	257	256	253	257	(42)	
(43)		Edh at Noon	31	42	55	65	63	61	55	52	48	41	33	29	(43)	
(44)	All-Sky Solar Radiation	RadAvg	836	1153	1586	1961	2208	2211	2119	1962	1698	1359	939	739	(44)	
(45)		RadStd	49	52	86	110	108	164	87	115	102	51	43	47	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+142	+65

Nomenclature: See separate page